

EDUCATION

Syracuse University: Master of Science in Computer Engineering

Anticipated Graduation May 2027

Syracuse University: B.S. in Computer Engineering (Minor: Physics)

Graduated May 2026 Magna Cum Laude | GPA: 3.60

Relevant Courses:

Signals & Systems Analysis * Embedded Systems & Microcontroller Lab * Robotics Programming Lab * Computer Architecture & VLSI Design * Computational Physics * Operating Systems Design * Object-Oriented Programming (C++) * Computer & Network Security * Database Management Systems

PROJECTS

Hockey Bot (Undergraduate Capstone)

- Designed and built an autonomous CoreXY gantry-based air hockey robot as a 4-person capstone team, demonstrated at ECS Open House 2026 including 3 difficulty levels featuring bounce-path trajectory prediction.
- Computer vision pipeline using OpenCV (background subtraction, HSV filtering, contour detection) on a Linux host; real-time NEMA-17 stepper motor control via Teensy 4.0 microcontroller over USB serial.

'Smart Copy Machine' – Junior Year Design Project

- Designed & built a gantry-style pen plotter utilizing a Raspberry Pi, stepper motors, driver chips, cameras, and OpenCV to scan and replicate hand drawn images by processing images into a series of coordinate-based motor commands.

Search and Rescue Robot

- Created a search and rescue robot that used ROS2 and AprilTags for localization for autonomous navigation around a room and through doors, scanning for a paper box that it would pick up & retrieve, returning to its start.

Custom Keyboard PCB

- Designed and assembled a fully custom keyboard PCB. Features USB-C, Bluetooth (Nice!Nano V2 MCU), ~5hr battery, ZMK firmware, rotary encoders, OLED display, and TRRS expansion port.

'Aperture Analysis' – Computational Physics Independent Study

- Built a C++ program using a custom FFT/IFFT implementation to process stellar images into coordinate data and reconstruct telescope aperture shapes from diffraction spike patterns. Used Python to visualize the results.

WORK EXPERIENCE

Syracuse University Student Machine Shop – Student Machinist (Aug 2023 – Present)

- Assisted students with machining tasks across lathes, mills, and 3D printers for coursework and independent projects. Ensured safe and proper operation of shop equipment and tooling.

AEC Engineering – Control Panel Technician (May 2025 – Sep 2025)

- Fabricated industrial control panels to UL508A standards. Ensured panel function with verification. Verification steps included panel ring-outs, functional testing, PLC logic testing and circuit debugging.
- Improved shop efficiency by developing the framework for a weight-based inventory tracking system for components and wire spools.

Strout's Point Wharf Company – Dock Staff (Jun 2019 – Aug 2023)

- Oversaw and led a team of ~8 other dock staff while performing typical duties, such as assisting customers and preventative maintenance of equipment. Coordinated dock space with mechanics to improve operational efficiency.

LEADERSHIP

Formula SAE - Citrus Racing, Syracuse University (Oct 2022 – Present)

Current Role: Lead of EV Development:

- Directing design, fabrication, programming efforts of a custom high-voltage battery pack, motor control, and various safety systems to transition the team's race car from internal combustion to the school's first electric vehicle powertrain.

Prior Roles: Head of Electronics, President, Chief Engineer

- **Chief Engineer:** Led the development cycle of the 2026 car, delivering the highest scoring Formula SAE vehicle for Syracuse University in the 2000s era at a 120-team competition.
- **President:** Managed \$50,000 annual team budget, secured corporate sponsorships, and oversaw financial and strategic planning with university administration. Grew the team from ~5 to over 40 active members through recruitment efforts and overall team leadership.
- **Lead of Electronics:** Designed and implemented all vehicle electrical systems, including custom wiring harnesses, ECU control software, and data acquisition systems.

Undergraduate Teaching Assistant – Physics 307 Science and Computers I (Aug 2025 – Dec 2025)

- Provided individualized support to students learning computational physics, guiding them in building accurate models with C and Python on Linux platforms.

SKILLS

Languages: C/C++, C#, Java, Python, Assembly/Machine Code; **Tools & Software:** Git, KiCad, Fusion 360, SPICE, OpenCV, ROS 2, Microsoft/Google Office; **Hardware:** PCB Design & bring up, Stepper Motor Systems, PLCs, Machine Shop (Lathe, Mill, 3D Printing) **OS:** Windows, Linux